Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 3H nr. 12.8

$r = 2 + \sin^2 4\theta$

1. Domein

   $\operatorname{dom} r = \mathbb{R}$

   $p = 4\theta \Rightarrow \theta = \dfrac{\pi}{4}$

   $\Rightarrow$ we onderzoeken op het beperkt domein $\left[0, \dfrac{\pi}{4}\right]$

2. Tekenonderzoek van $r(\theta)$

   $2 + \sin^2 4\theta = 0$

   $\sin^2 4\theta = 0$

   $\sin 4\theta = 0$

   $4\theta = k\pi$

   $\theta = \dfrac{k\pi}{4}$

   Maar omdat $\sin^2 4\theta \geq 0$ geldt:

   $r(\theta) = 2 + \sin^2 4\theta \geq 2$

   $\Rightarrow r^{(-1)}(0) = \varnothing$

θ	0		$\frac{\pi}{4}$
$r(\theta)$	2	$+$	2

3. Tekenonderzoek van $r'(\theta)$

   $r(\theta) = 2 + \sin^2 4\theta$

   $r'(\theta) = 2\sin 4\theta \cdot \cos 4\theta \cdot 4$

$r'(\theta) = 8\sin 4\theta \cos 4\theta$

$8\sin 4\theta \cos 4\theta = 0$

$\sin 4\theta = 0 \quad \text{of} \quad \cos 4\theta = 0$

$\sin 4\theta = 0 \Rightarrow 4\theta = k\pi \Rightarrow \theta = \dfrac{k\pi}{4}$

$\cos 4\theta = 0 \Rightarrow 4\theta = \dfrac{\pi}{2} + k\pi \Rightarrow \theta = \dfrac{\pi}{8} + \dfrac{k\pi}{4}$

$\Rightarrow r'^{(-1)}(0) = \left\{0, \dfrac{\pi}{8}, \dfrac{\pi}{4}\right\}$

θ	0		$\frac{\pi}{8}$		$\frac{\pi}{4}$
$r'(\theta)$	0	$+$	0	$-$	0

Maximum:

$r\left(\dfrac{\pi}{8}\right) = 2 + \sin^2\left(4 \cdot \dfrac{\pi}{8}\right)$

$r\left(\dfrac{\pi}{8}\right) = 2 + \sin^2\left(\dfrac{\pi}{2}\right)$

$r\left(\dfrac{\pi}{8}\right) = 2 + 1$

$\Rightarrow \max\ r\left(\dfrac{\pi}{8}\right) = 3$

Minimum (randpunten):

$r(0) = 2 + \sin^2(0)$

$r(0) = 2$

$r\left(\dfrac{\pi}{4}\right) = 2 + \sin^2\left(4 \cdot \dfrac{\pi}{4}\right)$

$r\left(\dfrac{\pi}{4}\right) = 2 + \sin^2(\pi)$

$r\left(\dfrac{\pi}{4}\right) = 2$

$\Rightarrow \min\ r(0) = \min\ r\left(\dfrac{\pi}{4}\right) = 2$

4. Schets

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$\Rightarrow$ Samenvattende tabel

θ	0		$\frac{\pi}{8}$		$\frac{\pi}{4}$
$r(\theta)$	$+$		$+$		$+$
$r'(\theta)$	0	$+$	0	$-$	0
$r(\theta)$	$m(2)$	$\nearrow$	$M(3)$	$\searrow$	$m(2)$

References